# AC Signals & Power

## Sinusoidal Signals

The general equation for a sinusoidal voltage is given as

$$v(t) = V_m \cos(\omega t + \vartheta)$$

where $\omega$ is the angular frequency is rad/s. v(t) also repeats across one period T, hence

$$\omega T = 2\pi \Rightarrow T = \frac{2\pi}{\omega}$$

The frequency of the signal is also inversely related to the period, as

$$T = \frac{1}{f}$$

Hence,

$$\omega = 2\pi f$$

Recall the RMS value for an AC signal, that is for an AC signal $s(t)$, its RMS value is

$$s_{rms} = \sqrt{\frac{1}{T}\int_0^T s^2(t)\ dt}$$

In the case of our general sinusoidal voltage,

$$v_{rms} = \frac{V_m}{\sqrt{2}}$$

This RMS value is also called the effective value, and is most used to represent an AC signal, however other metrics like peak to peak values $(2V_m)$, or peak values $(V_m)$ may be used too.

## Phasors

Another way to represent a sinusoidal signal is with a complex number, called a phasor, which encodes the amplitude and phase angle in a vector rotating in the complex plane.

In general, for a sinusoid $v(t) = V_m \cos(\omega t + \vartheta)$, its phasor is

$$\mathbf{v} = V_m e^{j\vartheta} = V_m \angle \vartheta$$

Notice how time is not encoded into the phasor, but if we were to encode time and angular frequency the phasor would be

$$\mathbf{v} = V_m e^{(\omega t + \vartheta)} = V_m \angle(\omega t + \vartheta)$$

and the real component is the same as the original signal.

Note that for signals in the form of $v(t) = V_m \sin(\omega t + \vartheta)$, it can be rewritten as

$$\begin{aligned} v(t) &= V_m \sin(\omega t + \vartheta) \\ &= V_m \cos\left(\omega t + \vartheta - \frac{\pi}{2}\right) \end{aligned}$$

and hence its phasor is

$$\mathbf{v} = V_m \angle\left(\vartheta - \frac{\pi}{2}\right) = V_m \angle\left(\vartheta - 90^\circ\right)$$

Phasors allow us to combine signals together easily. Say we had two VAC sources in series of

$$\begin{aligned} v_1(t) &= 20\cos(\omega t - 45^\circ) \\ v_2(t) &= 10\sin(\omega t + 60^\circ) \end{aligned}$$

The sum of the two signals will be the effective voltage of the two sources. To add them, we can convert them into their phasors:

$$\begin{aligned} \mathbf{v}_1 &= 20\angle -45^\circ \\ \mathbf{v}_2 &= 10\angle(60^\circ - 90^\circ) = 10\angle -30^\circ \end{aligned}$$

Then the sum $\mathbf{v}_1 + \mathbf{v}_2$ can be found as

$$\begin{aligned} \mathbf{v}_1 + \mathbf{v}_2 &= 20\angle -45^\circ + 10\angle -30^\circ \\ &= 20\cos(-45^\circ) + j20\sin(-45^\circ) + 10\cos(-30^\circ) + j10\sin(-30^\circ) \\ &= 20\cos(45^\circ) - j20\sin(45^\circ) + 10\cos(30^\circ) - j10\sin(30^\circ) \\ &= 20\cos(45^\circ) + 10\cos(30^\circ) - j20\sin(45^\circ) - j10\sin(30^\circ) \\ &= 10\sqrt{2} + 5\sqrt{3} - j(10\sqrt{2} + 5) \\ &= \sqrt{(10\sqrt{2} + 5\sqrt{3})^2 + (10\sqrt{2} + 5)^2}\angle \tan^{-1}\left(-\frac{10\sqrt{2} + 5}{10\sqrt{2} + 5\sqrt{3}}\right) \\ &= 29.77\angle -40.01^\circ \end{aligned}$$

which can be written as a sinusoidal expression as

$$v_{sum} = 29.77\cos(\omega t - 40.01^\circ)$$

Phasors can also tell us the relationship between two different sinusoidal signals. For example, consider the two phasors from earlier:

$$\mathbf{v}_1 = 20\angle -45^\circ$$
$$\mathbf{v}_2 = 10\angle -30^\circ$$

Notice how the phase angle for v2 is greater than v1's phase angle, and since both vectors rotate anticlockwise, we say that v2 leads v1 by 15 degrees, or that v1 lags v2 by 15 degrees.

From the sinusoidal signals, we would see that peaks of v2 appear first, before v1, by 15 degrees or $\frac{\pi}{12}$ radians.

Complex Impedances

Impedance is the ratio of the phasors of the voltage to the current, defined similarly to resistance.

For a resistor connected to a voltage source, the voltage and current will vary in phase with each other by Ohms law, and since they are in phase,

$$Z_R = \frac{V_m\angle\vartheta}{I_m\angle\vartheta}$$
$$= \frac{V_m}{I_m}\angle(\vartheta - \vartheta) = \frac{V_m}{I_m}\angle 0^\circ \quad \left[\frac{V_m}{I_m} \in \mathbb{R}\right]$$

hence the impedance of a resistor is a real number, equal to its resistance.

However, inductors and capacitors have a complex impedance.

For an inductor, the fundamental equation is

$$v(t) = Li'(t)$$

Say a sinusoidal current is supplied,

$$\mathbf{i} = I_m \cos(\omega t + \theta)$$

The current phasor is then

$$\mathbf{i} = I_m e^{j(\omega t + \theta)}$$

and hence

$$\frac{d\mathbf{i}}{dt} = \frac{d}{dt} I_m e^{j(\omega t + \theta)}$$
$$= \frac{d}{dt} I_m e^{j\omega t + j\theta}$$
$$= j\omega I_m e^{j\omega t + j\theta}$$
$$= j\omega \mathbf{i}$$

Hence the voltage across the inductor is

$$\mathbf{v} = Lj\omega\mathbf{i}$$

Thus,

$$Z_L = \frac{\mathbf{v}}{\mathbf{i}} = \frac{Lj\omega\mathbf{i}}{\mathbf{i}} = j\omega L$$

Notice how the impedance is purely imaginary, hence we can also call this the reactance of the inductor. If the inductor has some non zero resistance, then the inductance will have some real term. Thus, an inductor only has a purely complex impedance if it has purely inductance.

If we were to find an expression for the voltage phasor, it would be

$$\begin{aligned} \mathbf{v} &= Z_L\mathbf{i} \\ &= Lj\omega I_m e^{j(\omega t+\theta)} \\ &= L(e^{j\frac{\pi}{2}})\omega I_m e^{j(\omega t+\theta)} \\ &= L\omega I_m e^{j\left(\omega t+\theta+\frac{\pi}{2}\right)} \end{aligned}$$

Notice how voltage now has an extra $+\frac{\pi}{2}$ phase angle term, which means that the voltage will lead the current by $\frac{\pi}{2}$ radians, or 90 degrees.

We can apply a similar method for finding the impedance of a capacitor. First, the fundamental equation of a capacitor is

$$i(t) = Cv'(t)$$

Then, apply a AC voltage source of

$$v(t) = V_m \cos(\omega t + \theta)$$

and again, its phasor is

$$\mathbf{v} = V_m e^{j(\omega t+\theta)}$$

and its derivative is

$$\frac{d\mathbf{v}}{dt} = j\omega V_m e^{j(\omega t+\theta)} = j\omega\mathbf{v}$$

hence

$$\mathbf{i} = j\omega C\mathbf{v} \Rightarrow \mathbf{v} = \frac{1}{j\omega C}\mathbf{i}$$

Thus the impedance is

$$
\begin{aligned}
Z_c &= \frac{\mathbf{v}}{\mathbf{i}} = \frac{1}{j\omega C} \\
&= \frac{j}{j^2\omega C} = -\frac{j}{\omega C}
\end{aligned}
$$

Hence if the voltage supplied to a capacitor is $\mathbf{v} = V_m e^{j(\omega t+\theta)}$,

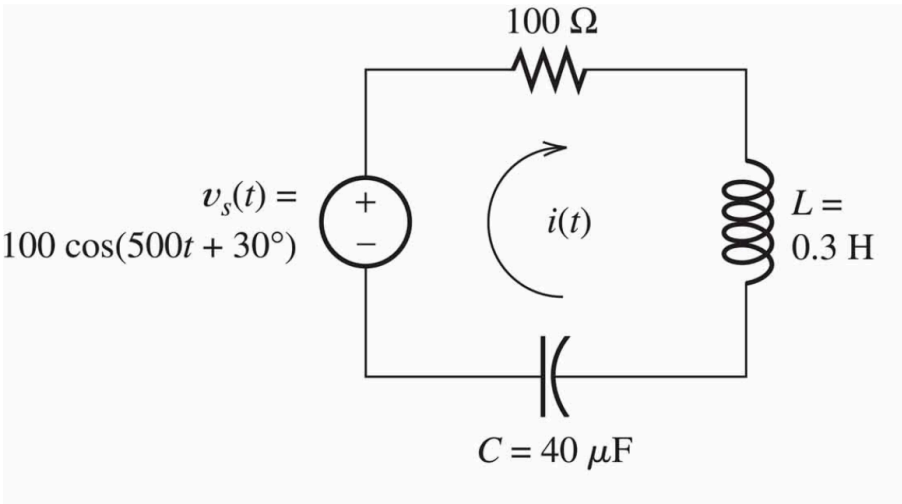
$$
\begin{aligned}
\mathbf{i} &= \frac{\mathbf{v}}{Z_C} \\
&= j\omega C V_m e^{j(\omega t+\theta)} \\
&= e^{j\frac{\pi}{2}} \omega C V_m e^{j(\omega t+\theta)} \\
&= V_m \omega C e^{j\left(\omega t+\theta+\frac{\pi}{2}\right)}
\end{aligned}
$$

which means that the current leads voltage by 90 degrees.

Circuit Analysis with Phasors & Complex Impedances

The general idea is that phasors can be treated as resistances, and hence KVL and KCL still applies.

Say we had to find the phasor voltage of each element in this series circuit:




Firstly, the phasor of the voltage source is

$$\mathbf{v} = 100\angle 30^\circ$$

We can easily find the impedances of each component, that is

$$
\begin{aligned}
Z_R &= 100\Omega \\
Z_L &= j\omega L = j(500)(0.3) = j150\Omega \\
Z_C &= -\frac{j}{\omega C} = -\frac{j}{500 \times 40 \times 10^{-6}} = -j50\Omega
\end{aligned}
$$

Notice how impedance has units Ohms.

Hence, the total impedance of the circuit is the sum of the impedances, that is

$$\begin{aligned} Z_{eq} &= Z_R + Z_L + Z_C \\ &= 100 + j150 - j50 \\ &= 100 + j100 \\ &= 100\sqrt{2}\angle 45^\circ \end{aligned}$$

From this, we can find the current phasor, as

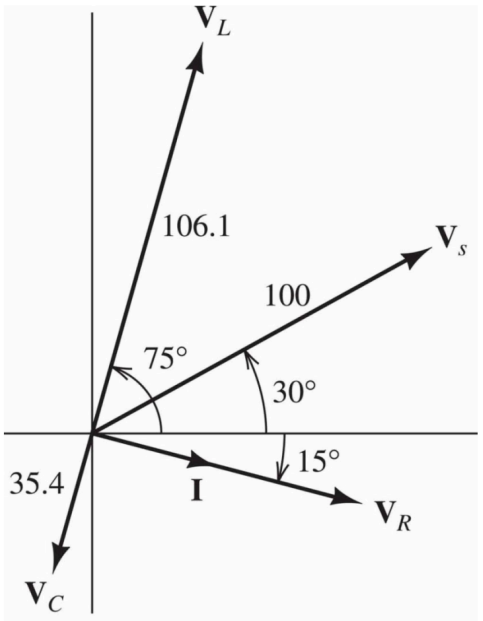
$$\begin{aligned} \mathbf{i} &= \frac{\mathbf{v}}{Z_{eq}} \\ &= \frac{100\angle 30^\circ}{100\sqrt{2}\angle 45^\circ} \\ &= \frac{100}{100\sqrt{2}}\angle(30^\circ - 45^\circ) \\ &= \frac{\sqrt{2}}{2}\angle -15^\circ \end{aligned}$$

Hence, to find the phasor voltage of each component in the circuit, we use $\mathbf{v} = \mathbf{i}Z$:

$$\begin{aligned} \mathbf{v}_R &= \left(\frac{\sqrt{2}}{2}\angle -15^\circ\right)(100) = 50\sqrt{2}\angle -15^\circ = 70.7\angle -15^\circ \\ \mathbf{v}_L &= \left(\frac{\sqrt{2}}{2}\angle -15^\circ\right)(150\angle 90^\circ) = 75\sqrt{2}\angle 75^\circ = 106.1\angle 75^\circ \\ \mathbf{v}_R &= \left(\frac{\sqrt{2}}{2}\angle -15^\circ\right)(50\angle -90^\circ) = 25\sqrt{2}\angle -105^\circ = 35.4\angle -105^\circ \end{aligned}$$

We can represent all the phasor voltages on a phasor diagram, which is just an Argand Diagram:

$\mathbf{I}$ = 0.707 $L$-15°
$\mathbf{V}_R$ = 70.7 $L$-15°
$\mathbf{V}_L$= 106.1 $L$75°
$\mathbf{V}_C$ = 35.4 $L$-105°
$\mathbf{V}_S$ = 100 $L$30°




which lets us see that the inductor's voltage leads the rest, followed by the source voltage, the resistor, then the capacitor.

One thing to note about complex impedances is that since they follow standard KVL and KCL laws, formulas derived from the follow too, thus parallel impedances can be found as

$$Z_{eq} = \left( \sum_{k=1}^{n} \frac{1}{Z_k} \right)^{-1}$$

Also,

$$\frac{1}{Z} = \frac{1}{|Z|\angle\theta} = \frac{1}{|Z|}\angle - \theta$$

AC Power

Consider a load that has the following voltage and current applied:

$$v(t) = V_m \cos(\omega t)$$
$$i(t) = I_m \cos(\omega t - \theta)$$

We can find the instantaneous power at a time t as

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t)\cos(\omega t - \theta)$$

We can split this into two terms, like so

$$
\begin{aligned}
p(t) &= V_m I_m \cos(\omega t)\left(\cos(\omega t)\cos(\theta) + \sin(\omega t)\sin(\theta)\right) \\
&= V_m I_m \cos(\omega t)\cos(\omega t)\cos(\theta) + V_m I_m \cos(\omega t)\sin(\omega t)\sin(\theta) \\
&= V_m I_m \cos^2(\omega t)\cos(\theta) + V_m I_m \cos(\omega t)\sin(\omega t)\sin(\theta) \\
&= V_m I_m \cos^2(\omega t)\cos(\theta) + \frac{V_m I_m}{2}\sin(2\omega t)\sin(\theta) \\
&= V_m I_m \left(\frac{1+\cos(2\omega t)}{2}\right)\cos(\theta) + \frac{V_m I_m}{2}\sin(2\omega t)\sin(\theta) \\
&= \frac{V_m I_m}{2}\cos(\theta)(1+\cos(2\omega t)) + \frac{V_m I_m}{2}\sin(2\omega t)\sin(\theta) \\
&= \frac{V_m I_m}{2}\cos(\theta) + \frac{V_m I_m}{2}\cos(\theta)\cos(2\omega t) + \frac{V_m I_m}{2}\sin(2\omega t)\sin(\theta) \\
&= \frac{V_m I_m}{2}\cos(\theta) + \frac{V_m I_m}{2}\left(\cos(2\omega t)\cos(\theta) + \sin(2\omega t)\sin(\theta)\right) \\
&= \frac{V_m I_m}{2}\cos(\theta) + \frac{V_m I_m}{2}\cos(2\omega t - \theta)
\end{aligned}
$$

where $\theta$ is the phase difference between V and I.

Since $\theta$ is constant, there is a constant term added to a sinusoidal term. This means that the power oscillates about an average power of $\frac{V_m I_m}{2}\cos(\theta)$, or

$$
\frac{V_m I_m}{2}\cos(\theta) = \frac{V_m I_m}{\sqrt{2}\sqrt{2}}\cos(\theta) = V_{rms} I_{rms}\cos(\theta)
$$

and oscillates by the term $\frac{V_m I_m}{2}\cos(2\omega t - \theta)$, which also shows that it oscillates at double the angular frequency.

Three Phase Power

Three phase power typically involves three sinusoidal voltages of the same magnitude but out of phase of each other, most typically 120 deg out of phase of one another. This setup is most frequently used as it is efficient; requiring little wiring, has a constant power output, and works with three phase AC motors, which have a non zero starting torque, making these actuators useful

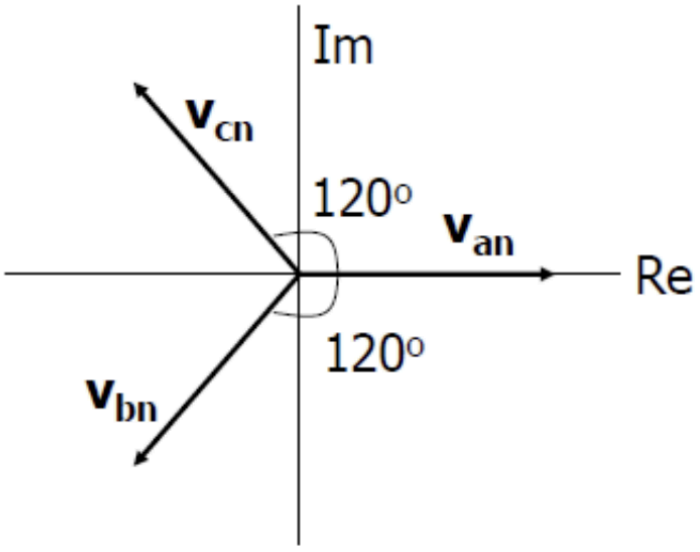
In general, the three expressions of the voltages are

$$
\begin{aligned}
v_{an}(t) &= V_Y \cos(\omega t) \\
v_{bn}(t) &= V_Y \cos(\omega t - 120^\circ) \\
v_{cn}(t) &= V_Y \cos(\omega t + 120^\circ)
\end{aligned}
$$

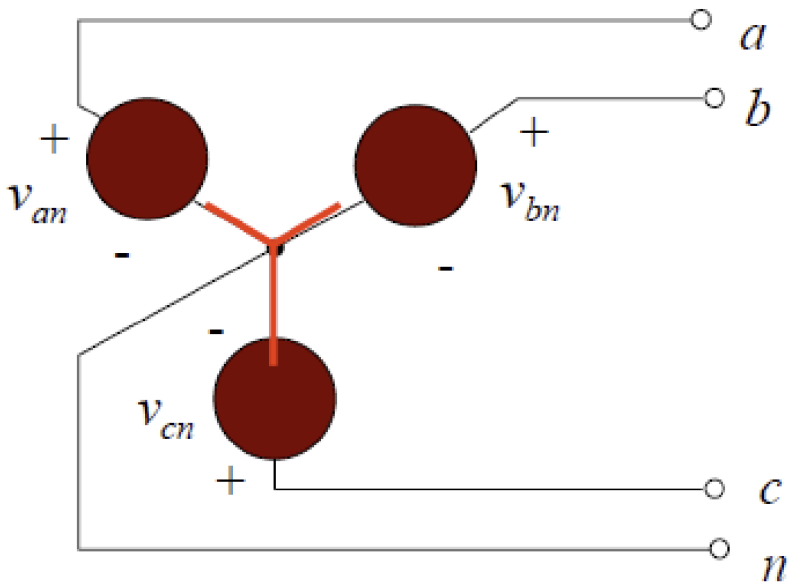
and hence the respective phasors are

$$
\mathbf{v}_{an} = V_Y\angle 0^\circ,\ \mathbf{v}_{bn} = V_Y\angle -120^\circ,\ \mathbf{v}_{cn} = V_Y\angle 120^\circ
$$

Notice how the sum of the phasors are equal to 0, which can also be seen on the phasor diagram:




Physically, to get a 3 phase power source we use wye connected voltage sources.

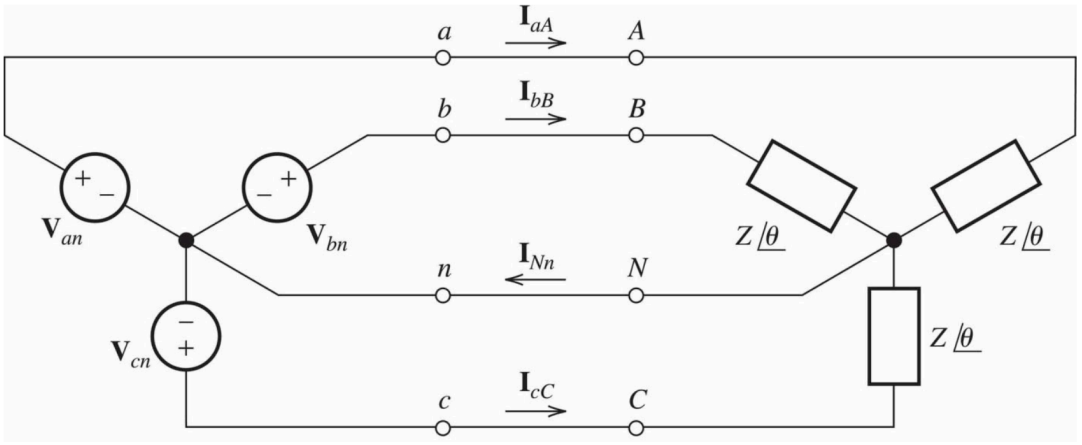



For this given setup, since a leads b and b leads c, we say this setup has a positive phase sequence, since the voltages reach the peaks in order abc.

If b and c were interchanged, the order would be acb, which has a negative phase sequence.

The phase sequence is useful as it can be used to change the direction of a 3 phase AC motor, by switching b and c.

Typically, the Wye connected power source is connected to wye connected loads:




The loads are usually equal, with the same impedance, thus called a balanced three phase load.

We can refer to each segment of the wye-wye connection as a phase, for example, phase B's source is $\mathbf{v}_{bn}$, or that $\mathbf{v}_{bn}$ is the phase voltage of B.

The phase voltages may also be called the line to neutral voltage, or the line voltage.

The current flowing from the voltage sources to the load is also called the line current.

Now, consider the line current for one of the phases, say phase A:

Since all the loads are the same, they have the same impedance in the form of $Z\angle\theta$.

Hence the line current is

$$\mathbf{I}_{aA} = \frac{\mathbf{V}_{an}}{Z\angle\theta} = \frac{V_Y\angle 0^\circ}{Z\angle\theta} = I_Y\angle -\theta$$

In general, the phasors for all three currents are

$$\mathbf{I}_{aA} = I_Y\angle -\theta,\ \mathbf{I}_{bB} = I_Y\angle -\theta - 120^\circ,\ \mathbf{I}_{aA} = I_Y\angle -\theta + 120^\circ$$

Notice how their sum is still 0, and since KCL states that $i_{nN} = i_{aA}(t) + i_{bB}(t) + i_{cC}(t)$, the neutral current is 0 and can be eliminated without affecting the rest of the circuit.

Another thing to note is that since there is a common neutral wire, less wiring is needed instead of having three neutral wires for each load.

For the power dissipated, we can find it as

$$p(t) = \sum_{k=1}^{3} v_{kK}(t) i_{kK}(t)$$
$$= \sum_{k=1}^{3} V_Y \cos(\omega t + \phi) I_L \cos(\omega t + \phi - \theta)$$

where $\phi$ is the phase angle for each phase, in other words, $\phi \in \{0, 120^\circ, -120^\circ\}$.

Hence, using $\cos(x)\cos(y) = \frac{1}{2}\left(\cos(x-y) + \cos(x+y)\right)$,

$$p(t) = \sum_{k=1}^{3} \frac{V_Y I_L}{2} \left(\cos(\omega t + \phi - (\omega t + \phi - \theta)) + \cos(\omega t + \phi + \omega t + \phi - \theta)\right)$$
$$= \sum_{k=1}^{3} \frac{V_Y I_L}{2} \left(\cos(\theta) + \cos(2\omega t + 2\phi - \theta)\right)$$
$$= \frac{V_Y I_L}{2} \sum_{k=1}^{3} \left(\cos(\theta) + \cos(2\omega t + 2\phi - \theta)\right)$$
$$\frac{V_Y I_L}{2} \left(3\cos(\theta) + \cos(2\omega t - \theta) + \cos(2\omega t - 240^\circ - \theta) + \cos(2\omega t + 240^\circ - \theta)\right)$$
$$\frac{V_Y I_L}{2} \left(3\cos(\theta) + \cancel{\cos(2\omega t - \theta) + \cos(2\omega t + 120^\circ - \theta) + \cos(2\omega t - 120^\circ - \theta)}\right)$$
$$= \frac{V_Y I_L}{2} \left(3\cos(\theta)\right)$$
$$= \frac{3V_Y I_L}{2} \cos(\theta)$$
$$= \frac{3V_Y I_L}{\sqrt{2}\sqrt{2}} \cos(\theta) = 3V_{rms} I_{rms} \cos(\theta)$$

again, $\theta$ is the phase angle difference between voltage and current.

Notice how this produces a constant power even though AC voltages were used, hence meaning the torque required and produced is constant, causing reduced vibrations.

Say we were to measure voltage across two different lines, like finding $\mathbf{v}_{ab}$, which is known as the line to line voltage.

We know that $\mathbf{v}_{ab} = \mathbf{v}_{an} - \mathbf{v}_{bn}$, hence

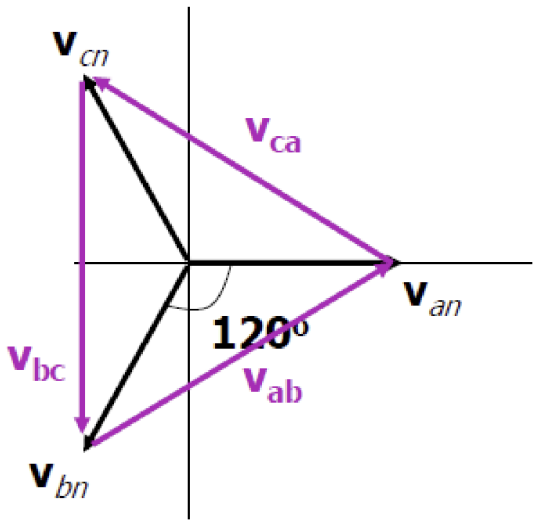
$$
\begin{aligned}
\mathbf{v}_{ab} &= V_Y\angle 0^\circ - V_Y\angle -120^\circ \\
&= V_Y\angle 0^\circ + (1\angle 180^\circ)(V_Y\angle -120^\circ) \\
&= V_Y\angle 0^\circ + V_Y\angle 60^\circ \\
&= V_Y + V_Y\cos(60^\circ) + jV_Y\sin(60^\circ) \\
&= V_Y + \frac{V_Y}{2} + jV_Y\frac{\sqrt{3}}{2} \\
&= V_Y\frac{3}{2} + jV_Y\frac{\sqrt{3}}{2} \\
&= V_Y\left(\frac{3}{2} + j\frac{\sqrt{3}}{2}\right) \\
&= V_Y\left(\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\angle\tan^{-1}\left(\frac{\sqrt{3}}{2} \div \frac{3}{2}\right)\right) \\
&= V_Y\left(\sqrt{\frac{9}{4} + \frac{3}{4}}\angle\tan^{-1}\left(\frac{\sqrt{3}}{2} \times \frac{2}{3}\right)\right) \\
&= V_Y\left(\sqrt{3}\angle\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)\right) \\
&= V_Y\sqrt{3}\angle 30^\circ
\end{aligned}
$$

In general, for $\mathbf{v}_{\alpha\beta}$ where $\alpha$ leads $\beta$,

$$\mathbf{v}_{\alpha\beta} = \mathbf{v}_{\alpha n} \times \sqrt{3}\angle 30^\circ$$

Note that this does not apply when the voltages are in a negative phase sequence.

On the phasor diagram, the line voltage is the vector connecting the voltage phasor vectors.




Consider the problem:

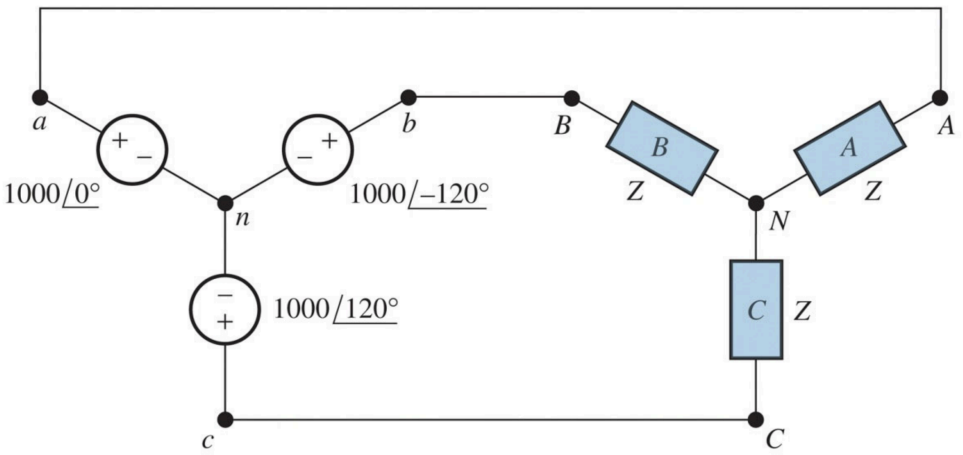
A balanced positive-sequence wye-connected 60-Hz three-phase source has line-to-neutral voltages of $V_Y = 1000$ V. This source is connected to a balanced wye-connected load. Each phase of the load consists of a 0.1-H inductance in series with a 50-Ω resistance. Find the line currents, the line-to-line voltages, the power, and the reactive power delivered to the load. Draw a phasor diagram showing the line-to-neutral voltages, the line-to-line voltages, and the line currents. Assume that the phase angle of $\mathbf{V}_{an}$ is zero.

▼ Solution

We are given that

$$V_Y = 1000V, f = 60Hz$$
$$L = 0.1H \ \& \ 50\Omega$$

and that the equivalent circuit is




We can first find the impedance of each load, that is

$$\begin{aligned} Z &= R + j\omega L \\ &= 50 + j(60 \times 2\pi)(0.1) \\ &= 50 + j12\pi \\ &= \sqrt{50^2 + (12\pi)^2} \angle \tan^{-1}\left(\frac{12\pi}{50}\right) \\ &= 62.62\angle 37.02^\circ \end{aligned}$$

Hence, for phase A, the line current is:

$$\begin{aligned} \mathbf{I}_{aA} &= \frac{\mathbf{V}_{an}}{Z} \\ &= \frac{1000\angle 0^\circ}{62.62\angle 37.02^\circ} \\ &= 15.97\angle -37.02^\circ \end{aligned}$$

similarly, for phases B and C,

$$\mathbf{I}_{bB} = 15.97\angle(-120^\circ - 37.02^\circ) = 15.97\angle -157.02^\circ$$
$$\mathbf{I}_{cC} = 15.97\angle(120^\circ - 37.02^\circ) = 15.97\angle -82.98^\circ$$

Hence, the line voltages can be found by $\mathbf{v}_{\alpha\beta} = \mathbf{v}_{\alpha n} \times \sqrt{3}\angle 30^\circ$, which are

$$\mathbf{v}_{ab} = \mathbf{v}_{an} \times \sqrt{3}\angle 30^\circ = (1000\angle 0^\circ)(\sqrt{3}\angle 30^\circ) = 1000\sqrt{3}\angle 30^\circ$$
$$\mathbf{v}_{bc} = \mathbf{v}_{bn} \times \sqrt{3}\angle 30^\circ = (1000\angle -120^\circ)(\sqrt{3}\angle 30^\circ) = 1000\sqrt{3}\angle -90^\circ$$
$$\mathbf{v}_{ca} = \mathbf{v}_{cn} \times \sqrt{3}\angle 30^\circ = (1000\angle 120^\circ)(\sqrt{3}\angle 30^\circ) = 1000\sqrt{3}\angle 150^\circ$$

And lastly, the power supplied to the loads is given as

$$P_Y = \frac{3V_Y I_L}{2}\cos(\theta) = 3V_{rms}I_{rms}\cos(\theta)$$

and since

$$\mathbf{V}_{an} = 1000\angle 0^\circ,\ \mathbf{I}_{aA} = 15.97\angle -37.02^\circ$$
$$\Rightarrow \theta = -37.02^\circ$$

the total power dissipated in all three loads is

$$P_Y = \frac{3}{2}(1000)(15.97)\cos(-37.02^\circ) = 19.126kW$$